

Mid-term examination

FULL NAME: _____

STUDENT ID: _____

Question	Max. points	Your points
1	10	
2	15	
3	8	
4	6	
5	8	
6	8	
Total	55	

- Attempt all questions. Questions may appear on either side of a sheet.
- The exam is **120 minutes** in duration. The maximum score is **55 points**.
- The use of a non-programmable scientific calculator and an A4 sheet with handwritten information on one side is allowed.
- Write your answers in this booklet. Request for extra paper only when needed.
- Return the extra sheets also when returning the exam
- Sign the SUTD honour code in the back of this page.



SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN HONOUR CODE

"As a member of the SUTD community, I pledge to always uphold honourable conduct. I will be accountable for my words and actions, and be respectful to those around me."

Introduction to the SUTD Honour Code

What is the SUTD Honour Code?

The SUTD Honour Code was established in conjunction with the school's values and beliefs, in good faith that students are able to discern right from wrong, and to uphold honourable conduct. It is an agreement of trust between the students and the staff and faculty of SUTD, and serves as a moral compass for students to align themselves to. Being in a university that aspires to nurture the leaders of tomorrow, it calls for students to behave honourably, not just solely in their academic endeavours, but also in everyday life.

What the Honour Code encompasses

Integrity & Accountability

To be honourable is to do what is right even when nobody is watching, and to be accountable for the things one does. One should always be accountable for one's words and actions, ensuring that they do not cause harm to others. Putting oneself in a favourable position at the expense of others is a compromise of integrity. We seek to create a community whereby we succeed together, and not at the expense of one another.

Respect

Part of being honourable is also respecting the beliefs and feelings of others, and to never demean or insult them. Should conflicts arise, the aim should always be to settle them in a manner that is non-confrontational, and try to reach a compromise. We will meet people of differing beliefs, backgrounds, opinions, and working styles. Understand that nobody is perfect, learn to accept others for who they are, and learn to appreciate diversity.

Community Responsibility

In addition to that, being honourable also involves showing care and concern for the community. Every individual has a duty to uphold honourable conduct, and to ensure that others in the community do likewise. The actions of others that display immoral or unethical conduct should not be condoned nor ignored. We should encourage each other to behave honourably, so as to build a community where we can trust one another to do what is right.

Student's signature

Question 1. (5 x 2 = 10 points) Let $(X_n : n \geq 0)$ be an irreducible, discrete time Markov chain with countably infinite state space I , probability transition matrix $P = ((p_{ij})), i, j \in I$. Considering this Markov Chain, circle around the right answer (TRUE or FALSE) for questions below. Ambiguous answers and unclear markings will result in zero points. You do NOT have to justify your answer for the objective questions in Parts (a) - (e) of Question Q1.

(a) It is possible for all states to be transient.

TRUE/ FALSE

(b) If the equation $\pi = \pi P$ has no solution then all states are transient.

TRUE/ FALSE

(c) If $p_{ii} = 0$ for all $i \in S$, then the chain must necessarily be periodic.

TRUE/ FALSE

(d) If $p_{ii} > 0$ for all $i \in S$, then a stationary distribution must exist for P .

TRUE/ FALSE

(e) An invariant distribution exists when the state space is $I := \{0, 1, 2, \dots\}$ and transition probabilities are given by,

$$p_{i,j} = \begin{cases} 1/2 & \text{if } i \geq 2 \text{ and } j \in \{i-1, i+1\}, \\ 1 & \text{if } i+j = 1. \end{cases}$$

TRUE/ FALSE

Question 2. ($7 + 7 + 1 = 15$ points) Smith is in jail and has 3 dollars; he can get out on bail if he has 8 dollars. A guard agrees to make a series of bets with him. If Smith bets A dollars, he wins A dollars with probability 0.4 and loses A dollars with probability 0.6. Find the probability that he wins 8 dollars before losing all of his money if,

- a) he bets 1 dollar each time (timid strategy);
- b) he bets, each time, as much as possible but not more than necessary to bring his fortune up to 8 dollars (bold strategy);
- c) which strategy gives Smith the better chance of getting out of jail?

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Question 3. (3 + 5 = 8 points) Suppose the vertices of a triangle are labelled 1, 2 and 3, and Tom (a cat) and Jerry (a mouse) move around these vertices in the following way. If Tom is at vertex i and Jerry is at vertex j , where $i \neq j$, then Tom will move to vertex j with probability $p \in (0, 1)$ and to the other vertex, say k , with probability $1 - p$. If Jerry sees Tom moving to vertex j , then Jerry instantaneously jumps to vertex k ; otherwise, Jerry remains at vertex j (so Tom never catches Jerry).

- a) Let X_n denote the location of Jerry after n movements of Tom. Does $(X_n : n \geq 0)$ form a Markov chain?
- b) Determine the probabilities of finding Jerry in Vertices 1, 2, and 3.

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Question 4 ($4 + 2 = 6$ points) A saleswoman flies around between Singapore, Bangkok, and Jakarta as follows. From Singapore she goes to Bangkok or Jakarta with probability $1/2$ each; from Bangkok she goes to Singapore with probability $3/4$ and to Jakarta with probability $1/4$; from Jakarta she always goes back to Singapore.

- a) What is the probability of finding the saleswoman in Singapore?
- b) What fraction of her trips are from Bangkok to Singapore?

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Question 5. (8 points) Suppose that $0 < p < q := 1 - p < 1$ for the discrete-time Markov chain model on the state-space $\{0, 1, 2, 3, \dots\}$ with transition probabilities given by $p_{0,1} = 1$, $p_{i,i+1} = p$ and $p_{i,i-1} = q$ for $i = 1, 2, \dots$. Find the expected time it takes for the chain to reach state '0' when the chain is begun at state '1' at time 0. ¹

¹Hint: Let k_i be the expected time it takes to reach state 0 when the Markov chain is initiated at state i . If you try to do first-step analysis, you will get an infinite system of linear equations. To solve the equations, you may use the symmetry in the problem to guess how k_i should behave as a function of i ? In particular, how many times you have to add k_1 to get k_2 ? How many times you have to add k_1 to get k_3 ?, and so on.

Question 6. (8 points) Let $(X_n : n \geq 0)$ be $\text{Markov}(\lambda, P)$. Suppose that we can observe the chain X_n only when it moves. So the observable process $(Z_m : m \geq 0)$ is given by

$$Z_m = X_{S_m},$$

where $S_0 = 0$ and $S_{m+1} = \min\{n \geq S_m : X_n \neq X_{S_m}\}$.

Is $(Z_m : m \geq 0)$ a Markov chain? Give reasons for your answer. Identify the transition probabilities if you argue it to be a Markov chain.

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